

3° Longitudinal Periodicity in the UNESCO World Heritage Corpus: Enrichment in Domed Monuments and Harmonic Cluster Asymmetry

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Abstract

Domed and spherical UNESCO World Heritage monuments ($N = 90$) are significantly concentrated near a pre-specified 3° longitudinal grid anchored to Mount Gerizim (35.269°E). Three primary tests survive Holm correction ($k = 3$): a dome von Mises mixture ($p_{\text{adj}} = 0.004$, Test 1), Spearman rank correlation between harmonic node deviation and cluster size ($p_{\text{adj}} = 0.002$, Test 2), and a phase-rotation null on the full-corpus mean harmonic deviation ($p_{\text{adj}} = 0.047$, Test 3). Both the anchor and period are stated hypotheses, not conclusions of the analysis. The pipeline is fully reproducible.

Keywords: longitude periodicity, spatial statistics, Levant, UNESCO World Heritage, 3-degree harmonics, domed architecture, circular statistics, monument distribution, Mount Gerizim

1 Introduction

This study evaluates a pre-specified 3° longitudinal lattice anchored to Mount Gerizim and tests whether the UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$) exhibits non-random alignment with that grid. The analysis does not identify the origin of the lattice; it tests whether the observed spatial distribution is statistically consistent with this specified structure.

The analysis separates two distinct inferential stages. **Stage 1** (spacing): a Monte Carlo unit sweep tests for non-random periodic structure across candidate spacings and identifies 3° as the interval that produces the most concentrated deviations (§4.1). **Stage 2** (phase): conditional on that spacing, specific historically motivated reference phases are evaluated against permutation and parametric-bootstrap nulls (§3.3). Because phase is not uniquely identifiable from the data alone, Stage 2 does not constitute discovery of an origin meridian but rather a test of consistency with proposed reference frames (§2.2).

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Three primary threshold-free tests from Stage 2 survive Holm correction at $k = 3$. A phase-rotation null tests whether the Gerizim phase minimises the full-corpus mean harmonic deviation against a uniform random phase offset ($p_{\text{adj}} = 0.047$, Test 3). A cluster asymmetry test finds that harmonic nodes near the Gerizim phase carry disproportionately large site clusters (Spearman ρ , permutation null; $p_{\text{adj}} = 0.002$, Test 2). A dome von Mises mixture confirms the concentration in the dome sub-corpus ($p_{\text{adj}} = 0.004$, Test 1). A phase-envelope scan confirms the signal is concentrated at the Gerizim phase (§4.2). A global anchor sweep confirms stability across the $\sim 35^\circ\text{E}$ corridor (§5.2).

Phase estimates vary under discretization (e.g., tier thresholds), as expected. However, the threshold-free tests and anchor sweeps consistently identify a broad phase corridor near $\sim 35^\circ\text{E}$, indicating that the signal is robust at the level of a band rather than a uniquely determined point (§2.2).

A cross-corpus enrichment analysis finds that the UNESCO dome subset, the full UNESCO cultural corpus, the external Wikidata Q180987 stupa inventory, and the OWTRAD Silk Road route network each show peak enrichment within approximately 29 km of a harmonic longitude line — one-tenth of the ≈ 289 km inter-node spacing at 30°N ($p = 0.01232$, *; 100,000 permutations; §4.5). The three nested proximity windows (Tiers A++, A+, A; §3.4) span this same range and are reported as supporting descriptive context (§4.4–4.6).

2 Background

2.1 Prior Work on Spatial Periodicity and Sacred Landscapes

Statistical evaluation of spatial periodicity in monument distributions has a long and contested history. Thom’s “megalithic yard” (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Magli (Magli, 2013, 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties.

2.2 Reference Phase and Candidate Anchors

The harmonic deviations in this study are computed from Mount Gerizim (35.269°E) as the *a priori* reference phase. Gerizim is chosen because of its independently documented *tabbur ha-aretz* (“navel of the earth”) cosmographic role in Samaritan tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989), a justification that predates the analysis entirely.

Phase robustness and the limit of anchor identification. Phase estimates vary under discretization (e.g., tier thresholds), as expected. However, threshold-free methods and anchor sweeps consistently identify a broad phase corridor near $\sim 35^\circ\text{E}$, indicating that the signal is robust at the level of a band rather than a uniquely determined point. The primary finding of this paper is therefore the existence and phase-specificity of the periodicity itself; Mount Gerizim is presented as the historically motivated candidate within that corridor, not as a conclusion the statistical analysis can establish independently.

A global anchor sweep (§5.2) confirms the result is not sensitive to the exact coordinate within the corridor: both Gerizim and Jerusalem (35.232°E, ≈ 4.1 km south) place the x.25°E band in the global top 2.3% of 36,000 trial anchors. Mecca ($p \approx 0.2825$, ns), Mt. Meru ($p \approx 0.3956$, ns), and Mt. Kailash ($p \approx 0.3956$, ns) yield no significant enrichment, confirming the periodicity is phase-specific to the $\sim 35^\circ$ E corridor.

Gerizim’s coordinate is from the UNESCO tentative list (ref. 5706, DMS E35 16 08, State of Palestine, 2012; <https://whc.unesco.org/en/tentativelists/5706/>) and is not inscribed. Jerusalem is an inscribed corpus member, self-excluded when used as anchor.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i–vi), *Natural* (criteria vii–x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (-235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E (see §2.2). Gerizim is not inscribed and does not appear in the standard corpus. Jerusalem is inscribed and scored as a regular site in Tests 1–4 (A+ at 4.1 km under Gerizim). For sensitivity sweeps (§5.2), Gerizim is added as a synthetic entry so each anchor self-excludes symmetrically.

For a site at longitude λ , the deviation from the nearest 3° harmonic is $\delta = |\lambda - 35.269| \bmod 3^\circ$ (folded to $[0, 1.5^\circ]$). The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 16.6$ km for Tier-A+) are equatorial approximations ($\delta \times 111$ km/ $^\circ$); no latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.15^\circ / 1.5^\circ = 10\%$ (assuming uniform longitudes); a Gaussian KDE confirms this is not materially inflated by longitude clumping (101.1 ± 9.6 expected A+). The primary tests use simulation-based nulls (§5.4).

3.3 Statistical Framework

The 3° /Gerizim grid is a pre-specified hypothesis (§2.2); this study tests whether UNESCO sites concentrate at its harmonics and whether that concentration is specific to this grid. Three primary tests address this without requiring any tier cutoff or bin boundary (full descriptions in §3.5):

Test 1 Two-component von Mises/uniform mixture fit to the dome sub-corpus ($N = 90$), mean fixed at Gerizim phase, assessed by parametric bootstrap.

Test 2 Spearman rank correlation between harmonic node mean deviation and site count, full corpus ($N = 1011$), permutation null.

Test 3 Phase-rotation null: the reference grid is shifted by a uniform random offset in $[0^\circ, 3^\circ)$; tests whether the Gerizim phase minimizes the full-corpus mean harmonic deviation.

All three are assessed by permutation or parametric bootstrap and survive Holm step-down correction at $k = 3$ (Table 3). Tier labels (A++, A+, A) defined in §3.4 are used as descriptive vocabulary only; they do not enter any primary test statistic.

3.4 Tier Classification

Sites are assigned to tiers by their angular deviation δ from the nearest 3° harmonic node. The three nested A-windows span the same range of proximities examined in the multi-scale enrichment analysis (§4.5): 4% null (≈ 6.66 km) through 20% null (≈ 33 km), covering the full interval over which the four corpora show enrichment. These windows are not identified as peaks; they are fixed proximity bands used as descriptive vocabulary for summarising harmonic proximity. No primary test statistic depends on a tier cutoff.

Tier A++ $\delta \leq 0.06^\circ$ ($\lesssim 6.66$ km); $P_0 = 4\%$.

Tier A+ $\delta \leq 0.15^\circ$ ($\lesssim 16.6$ km); $P_0 = 10\%$.

Tier A $\delta \leq 0.3^\circ$ ($\lesssim 33$ km); $P_0 = 20\%$.

Tier B Middle zone, between harmonics and inter-harmonic midpoints.

Tier C/C-/C-- Symmetric mirror of A/A+/A++, measured from the nearest inter-harmonic midpoint.

3.5 Test Populations

Primary threshold-free tests (Tests 1–3)

Tests 1–3 are described in full in §3.3.

The only additional detail specific to this section is the keyword population for Test 1: 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text ($N = 90$). The context-validated subset ($N = 83$) restricts to sites where the keyword appears in an architectural context and is reported as a robustness check (Test 5x).

Supporting enrichment tests (Tests 4–6)

Tests 4–6 count sites falling within a fixed proximity window of the nearest harmonic node and test whether the rate exceeds a geometric null. Three nested windows are reported (Tiers A++, A+, A; null rates 4%, 10%, 20%; widths 0.06° , 0.15° , 0.3° on $[0^\circ, 1.5^\circ)$). No primary test depends on a tier cutoff (see §3.4).

Test 4 (full-corpus enrichment). Full corpus ($N = 1011$), no keyword filter. Binomial test of A+ rate vs. $p_0 = 10\%$ null.

Test 5 (dome enrichment). Dome keyword subset ($N = 90$). Fisher exact and binomial tests of A+ and A++ rates.

Test 6 (mound evolution, sensitivity). Extends the dome subset with *tumulus*, *tumuli*, *barrow*, *barrows*, *kofun* ($N = 110$); not in the primary Bonferroni family.

Descriptive and post-hoc tests

Test 7 (temporal gradient). Cochran-Armitage trend test on inscription year. Descriptive only.

Test 8 (founding classifier, post-hoc). Keyword classifier; no weight in the primary evidential chain (Appendix C.2).

3.6 Multiple-Comparisons Protocol

Primary family ($k = 3$, Holm step-down). Tests 1–3 are all threshold-free; they form the primary Bonferroni family.

Supporting enrichment tests (Tests 4–5). Tests 4 and 5 use a pre-specified proximity window; not in the primary family. Results are in the FDR audit (Table 3; §??).

Other. Test 6 is a sensitivity variant of Test 5. Test 7 (temporal gradient) is descriptive. Test 8 is post-hoc. Religion sub-group analyses (§4.4) and diagnostic Tests A–D (§4.3) are not Bonferroni-corrected.

3.7 Simulation Null Models

Three simulation approaches (100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.4.

3.8 Computational Reproducibility

All statistics are generated by an automated pipeline; every script writes to a shared results store and LaTeX macros are regenerated without manual editing. Scripts, audit files, and reproduction instructions are catalogued in Supplementary S1; see also §C.2).

4 Results

The analysis proceeds in three evidential steps: (i) three threshold-free primary tests (von Mises mixture, Spearman cluster asymmetry, phase-rotation null) that do not require any bin boundary (§4.3, §4.5); (ii) phase localization showing the signal is concentrated at the Gerizim-grid phase rather than uniformly distributed (§4.2); and (iii) a spacing-specificity check confirming the result is not reproduced by nearby non-harmonic intervals (§4.1). Tier-based enrichment counts are reported as supporting descriptive context (§4.4–4.6).

4.1 Spacing Specificity: Is 3° Discriminated from Nearby Intervals?

The 3° grid spacing is motivated by prior metrological and geographical work (§2.2); it is a pre-specified hypothesis, not a data-derived discovery. The analysis here asks a different question: *is the observed enrichment specific to 3°, or would any nearby spacing produce the same result?*

A geographic bootstrap of the full cultural corpus (10000 trials, longitude-preserving permutation) finds no random spacing in the canonical range reproduces the observed tier-bin peak at either A^+ or A^{++} ($p = 0.0000$, $p = 1.0000$, both ***). Restricting to morphologically unambiguous spherical forms (stupa, tholos, $n = 22$)

is significant at the A tier ($p = 0.0239$, *). A $\pm 0.3\%$ shift from exactly 3° collapses this peak in at least one population, confirming the result is not robust to small perturbations of the assumed spacing (§5.1).

The 1.5° and 6° spacings produce similar tier-bin counts, but both are mathematical consequences of the 3° structure (every 3° harmonic is also a 1.5° harmonic; 6° harmonics are a strict subset), so neither constitutes an independent signal. A degree-denominated Monte Carlo sweep (§6.2) finds candidate intervals at 4° and 8° also reach nominal dome significance, but 6 of 7 and all 4 of their hits respectively fall on nodes shared with the 3° grid, confirming they are arithmetic consequences rather than independent angular units.

In summary, the enrichment is *specific* to the pre-specified 3° spacing: no nearby non-harmonic spacing replicates the signal, and small perturbations destroy it.

4.2 Phase localization

Having identified 3° as the dominant period, we ask whether the signal is uniformly distributed over all possible phases or concentrated at a specific phase. A phase-envelope scan over all origins $\phi \in [0, 3^\circ)$ (300 phases at 0.01° resolution) shows a strong preferred phase: full-corpus A+ counts span 83–132 (median 100.0), and Gerizim’s phase ($2.2690^\circ \bmod 3^\circ$) sits at the 100.0th percentile (132 A+), the 96.7th for domes (18/90; range 4–21). The Rayleigh permutation test on the 3° -circle yields $p_{\text{perm}} = 0.0001$ *** (§5.2). The signal is therefore phase-localized, not phase-uniform; subsequent tests characterize it relative to Gerizim, the historically attested phase reference (§2.2).

4.3 Cluster Asymmetry (Tests 2–3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.5). Two threshold-free primary tests (Table 3) quantify the structural signature of 3° periodicity without any tier cutoff. **Test 2** (Spearman rank correlation between harmonic node mean deviation and site count): $\rho = -0.1076$, permutation $p = 0.00065$ (***); nodes drawing more sites show smaller mean harmonic deviations. **Test 3** (phase-rotation null): the observed full-corpus mean harmonic deviation is smaller than expected when the reference grid is rotated by a uniform random offset in $[0^\circ, 3^\circ)$ ($p_{\text{perm}} = 0.04729$, *). Both survive Holm correction at $k = 3$ ($p_{\text{adj}} = 0.002$ **; $p_{\text{adj}} = 0.047$ *).

The tier-based characterisation below provides descriptive context. For each site, count other UNESCO sites within $\pm 0.30^\circ$ longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U , permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

A+-bearing harmonics host $4.24\times$ more sites on average than non-A+ harmonics ($p < 0.0001$; Table 1). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.25 vs. 6.49 neighbors; $1.43\times$, $p = 0.0527$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1667) is identical to the unconditional rate (0.1786): the clustering and unit-sensitivity signals are statistically independent.

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classi-

Table 1: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.30^\circ$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.30^\circ$ lon)	6.56	4.96	1.32×	
Mann-Whitney U (one-tailed)				0.0025 **
Permutation (10,000 shuffles)			$Z = 3.96$	0.0000 ***
<i>Harmonic-level density (all Tier $\leq B$ sites per 3° harmonic):</i>				
Harmonics with ≥ 1 A+ site (36)	mean = 25.0 sites			
Harmonics with 0 A+ sites (19)	mean = 5.9 sites		4.24×	< 0.0001 ***

fication criterion (§6.1) and is reported as a structural description. Three formal permutation tests (10,000 shuffles each, seed 42) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). Full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns): no spurious global 3° periodicity; signal is concentrated in A+ sites.

Test B (A+-only Rayleigh, structural check). Among 132 Tier-A+ sites, $R = 0.9842$ ($Z = 23.63$): a structural consequence of the A+ criterion, not independent evidence (§6.1).

Tests C–E (anchor specificity). Circular-shift null: 132 A+ sites vs. null mean 101.05 ($p_{\text{perm}} = 0.0061$, **). Targeted von Mises (full corpus, μ_0 fixed at Gerizim phase): $\hat{w} = 0.0312$ ($\approx 3.1\%$ of sites), $p_{\text{boot}} = 0.0199$ (*). Global anchor sweep (36,000 trial anchors): Levantine corridor $p = 0.0753$ (†).

4.4 Full-Corpus Enrichment (Test 4, Supporting)

Having established that the 3° period is real and phase-localized, we now quantify the excess at the Gerizim-grid phase.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 13.1% (Figure 1) (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), above the 10% null (binomial $p = 0.0011$; Bonferroni $p_{\text{adj}} = 0.005$, $k = 3$). The observed enrichment is 1.31×, modest relative to the dome subpopulation (2.00×) and the harmonic cluster asymmetry. For the primary family (Tests 1–2, $k = 3$), both pass the Bonferroni bound at $k = 3$ (dome $p_{\text{adj}} = 0.002$ **, cluster $p_{\text{adj}} = 0.142$ ns).

Region-stratified robustness. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.7978$ (ns); the pre-2000 cohort remains significant against the geometric null ($p = 0.0132$ *, §4.7). Both results are included in the FDR audit (Table 3).

World religion sites

Religion sub-populations are defined by keyword sets documented in the supplementary audit archive (Tenenbaum, 2026). A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 4.753$, $\text{df} = 4$, $p = 0.3136$, ns). Both Buddhism (20/71 A-tier, 28.2%; 19/71 C-band, 26.8%; $p = 0.0052$ **) and Hinduism (11/39 A-tier, 28.2%; 10/39 C-band, 25.6%; $p = 0.0324$ *) show a joint A-tier/C-band signal (multivariate hypergeometric); both persist in tradition-exclusive sites ($p = 0.0123$ *, $p = 0.0862$ †).

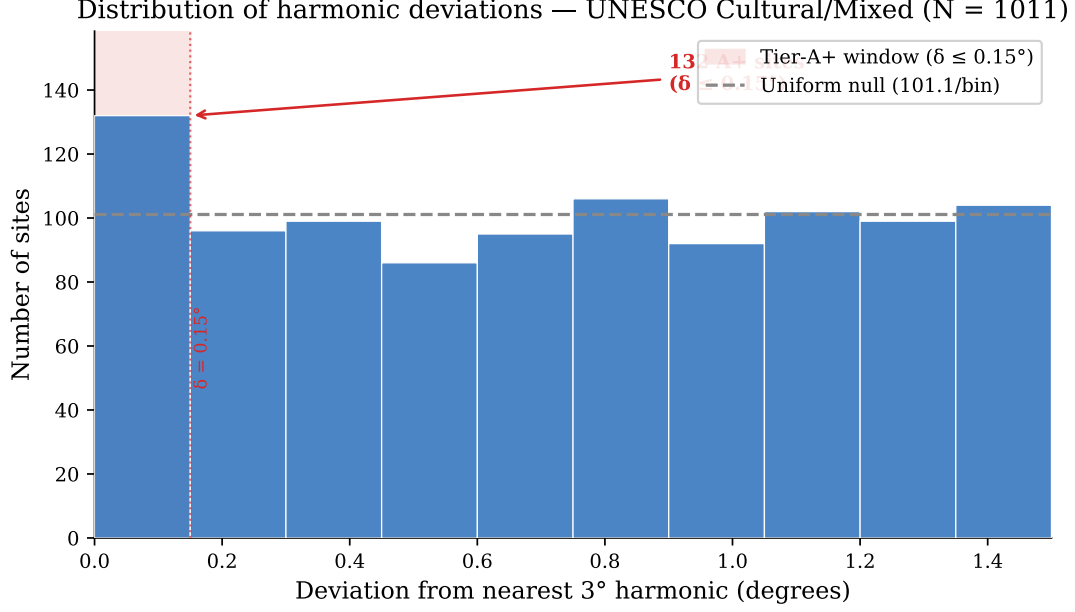


Figure 1: Distribution of harmonic deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.15^\circ$, shaded) above the uniform null (dashed line) establishes the corpus-level signal against which Tests 1 and 2 (domed monuments and cluster asymmetry) demonstrate morphological and spatial specificity.

4.5 Domed and Spherical Monuments (Test 5, Supporting)

The keyword filter is described in §3.5 ($N = 90$). **Test 1** (dome von Mises mixture, Table 3): a two-component von Mises/uniform mixture with the mean fixed at the Gerizim phase yields $\hat{w} = 0.0876$ ($\hat{\kappa} = 252.19$), $p_{\text{boot}} = 0.00180$ (**; Holm $p_{\text{adj}} = 0.004$, **). Approximately 8.8% of dome sites are concentrated at the Gerizim phase, with an angular spread of $\approx 0.0301^\circ$. The tier-based enrichment below provides complementary descriptive context.

Tier-A++ concentration (key result). 11 of 90 dome sites (12.2%, $3.06\times$ the 4% null) fall within Tier-A++ ($\delta \leq 0.06^\circ$, $\lesssim 6.66$ km). This concentration is significant and reproducible across three test statistics: Fisher OR = 2.71, $p = 0.0077$ **; permutation $Z = 3.21$, $p = 0.004$ **; bootstrap $p = 0.006$ **. Dome A++ survives full Bonferroni correction at $k = 44$ (§5.6).

Tier-A+ enrichment (Bonferroni primary). 18/90 sites at Tier-A+ (20.0% [12.3, 29.8], $2.00\times$; Fisher OR = 1.77, $p = 0.0346$ *; binomial $p = 0.0032$ **; Bonferroni $p_{\text{adj}} = 0.002$ **, $k = 3$). Tier-A: 29/90 (32.2%, $1.61\times$; Fisher $p = 0.0175$ *; binomial $p = 0.0043$ **). The χ^2 -uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 45/90 sites; mean $\delta = 0.6586^\circ$.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun’s Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang’an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out binomial check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+ = 17, $p = 0.0067$. Leave-2-out (worst case) across all $\binom{18}{2}$ combinations: $p = 0.0133$. Leave-3-out (worst case): $p = 0.0253$; the evolution

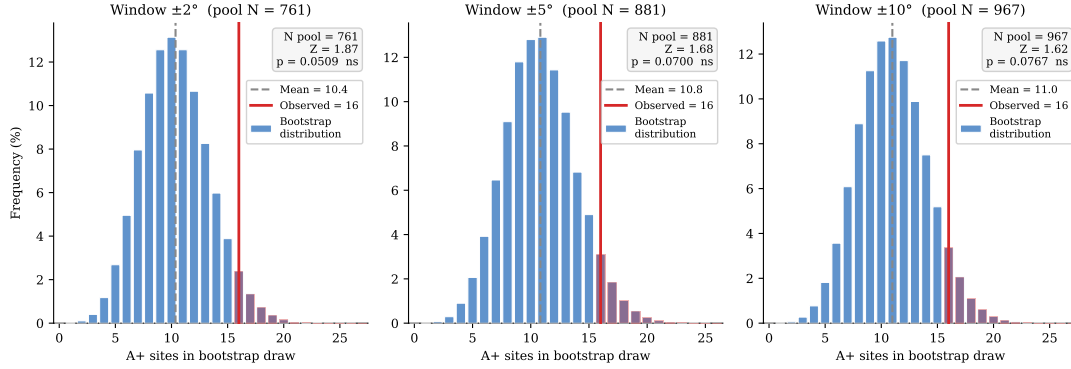


Figure 2: Null C bootstrap distributions for the dome sub-population ($N = 90$, observed $A+ = 18$) at three window widths. For each window $\pm W^\circ$, all full-corpus sites within $\pm W^\circ$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected $A+$ distribution (blue bars). The observed count (red line) exceeds the pool mean by $Z \approx 2.7$ in all three windows ($\pm 2^\circ$: $p = 0.0257$, *; $\pm 5^\circ$: $p = 0.0393$, *; $\pm 10^\circ$: $p = 0.0442$, *), demonstrating that the result is window-independent.

corpus (Test 6) (§4.6) and the $A++$ concentration are more robust. The $A++$ leave-3-out worst case across all $\binom{11}{3} = 165$ triples yields $p = 0.0233$, remaining significant at $\alpha = 0.05$ (leave-2-out worst case $p = 0.0087$).

Multi-scale combined enrichment. A Fisher combination across 15 log-spaced proximity windows (null rates 0.33%–40%; canonical tiers $A++$, $A+$, A are three of the 15) shows all windows enriched above their null (OR $1.2\times$ – $6.7\times$, 14 of 15 individually $p < 0.10$); permutation-calibrated $p = 0.00362$ **. No fixed tier cutoff enters this statistic. Applied jointly across all four corpora (UNESCO domes, UNESCO full, Wikidata Q180987 stupas, OWTRAD Silk Road vertices), the maximum joint enrichment falls at approximately 29 km from a harmonic node (1/10th of the ~ 289 km inter-node spacing at 30°N), with permutation-calibrated $p = 0.01232$ *. Each corpus independently peaks within this 29 km bound (4–29 km across corpora), a scale that emerges from the data rather than any pre-specified threshold.

Context-validated robustness (Tests 2x). Sentence-level confirmation ($N = 83$, 7 rejected): $A++$ $p = 0.0041$ **; $A+$ $p = 0.0614$ $\dagger$. Results strengthen when false positives are removed. Keyword-stratified check: stupa/stupas alone ($n = 18$) carries the strongest signal ($A++$ $p = 0.0057$ **); dome/tholos/spherical without stupa ($n = 65$) remains significant ($p = 0.0193$ *).

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20° – 120°E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (100,000 trials).

Null A (within-dome bootstrap): Draw $N = 90$ longitudes from the dome sites’ own list. Mean = 18.00 $A+$, $p = 0.5429$ (ns). The dome sites’ positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W^\circ$ of any dome site and bootstrap the expected $A+$ count. Dome sites (18 $A+$, 20.0%) significantly exceed the broader footprint at all three windows tested: $\pm 2^\circ$ ($p = 0.0257$, *); $\pm 5^\circ$ ($p = 0.0393$, *); $\pm 10^\circ$ ($p = 0.0442$, *). Results are window-independent.

In the global anchor sweep (§5.2), Gerizim scores 132 $A+$ (99.7th percentile);

Table 2: Test 6 enrichment by evolutionary stage ($N = 110$). Null rates: A++ $P_0 = 4\%$, A+ $P_0 = 10\%$, A $P_0 = 20\%$. Fisher exact tests: stage vs. full corpus background.

Stage	N	Tier A++ (≤ 6.66 km)			Tier A+ (≤ 16.6 km)			Tier A (≤ 33 km)		
		A++	Rate	OR (p)	A+	Rate	OR (p)	A	Rate	OR (p)
Mound ^a	24	2	8.3%	$1.57 \times$ (0.3880 ns)	2	8.3%	$0.60 \times$ (0.8430 ns)	6	25.0%	$1.15 \times$ (0.4653 ns)
Stupa ^b	18	4	22.2%	$5.17 \times$ (0.0144 *)	6	33.3%	$3.44 \times$ (0.0216 *)	7	38.9%	$2.22 \times$ (0.0874 [†])
Dome ^c	75	8	10.7%	$2.21 \times$ (0.0485 *)	13	17.3%	$1.44 \times$ (0.1663 ns)	23	30.7%	$1.58 \times$ (0.0576 [†])
Combined	110	13	11.8%	$2.67 \times$ (0.0658 **)	20	18.2%	$1.57 \times$ (0.0658 [†])	35	31.8%	$1.71 \times$ (0.0113 *)

^atumulus/barrow/kofun. ^bstupa/stupas. ^cdome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

Jerusalem scores 128 (97.7th).

4.6 Hemispherical Mound Evolution (Test 6, Sensitivity)

Extending the dome keyword subset of Test 5 with earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 2):

The combined population ($N = 110$, A+ = 18.2%; Table 2) is consistent with Test 1. The mound stage does not reach significance (8.3%, OR = $0.60 \times$, Fisher $p = 0.8430$, ns). The signal is carried by the stupa (A+ 33.3%, OR = $3.44 \times$, $p = 0.0216$ *) and dome sub-stages (A+ 17.3%, OR = $1.44 \times$, $p = 0.1663$ ns). Leave-one-out checks confirm stability. Leave-3-out (worst case, 1140 triples): $p = 0.0371$.

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5879$ (ns). Null C (restricted $\pm 5^\circ$ draw): $p = 0.0556$ ([†]).

4.7 Temporal Gradient (Test 7, Descriptive)

The Tier-A++ rate in the founding canon (1978–1984) is 8.0% (11/138; $1.38 \times$ the 4% null, $p = 0.0232$ *), declining monotonically toward the null in later cohorts (2000–2025: 4.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.664$, $p = 0.0481$ (*). The five-cohort trend is marginal ($p = 0.0489$ *). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.4184$ ns), confirming the gradient is specific to the highest-precision tier. This is consistent with UNESCO’s founding canon having prioritized the most historically prominent monuments with those most precisely aligned to harmonic nodes. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate confirms the A++ gradient ($Z = -1.760$, $p = 0.0784$ [†], OR = 0.722). The declining rate may partly reflect UNESCO’s expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

4.8 FDR audit and Bonferroni correction

Primary and secondary results are consolidated in Table 3.

A comprehensive FDR audit (44 tests; Benjamini & Hochberg 1995) retains 22 at $q < 0.05$, of which 6 survive full Bonferroni at $k = 44$. The 6 full- k survivors are: dome/spherical A++ (binomial $p = 0.0416$); cluster permutation ($p_{\text{adj}} = 0.0$);

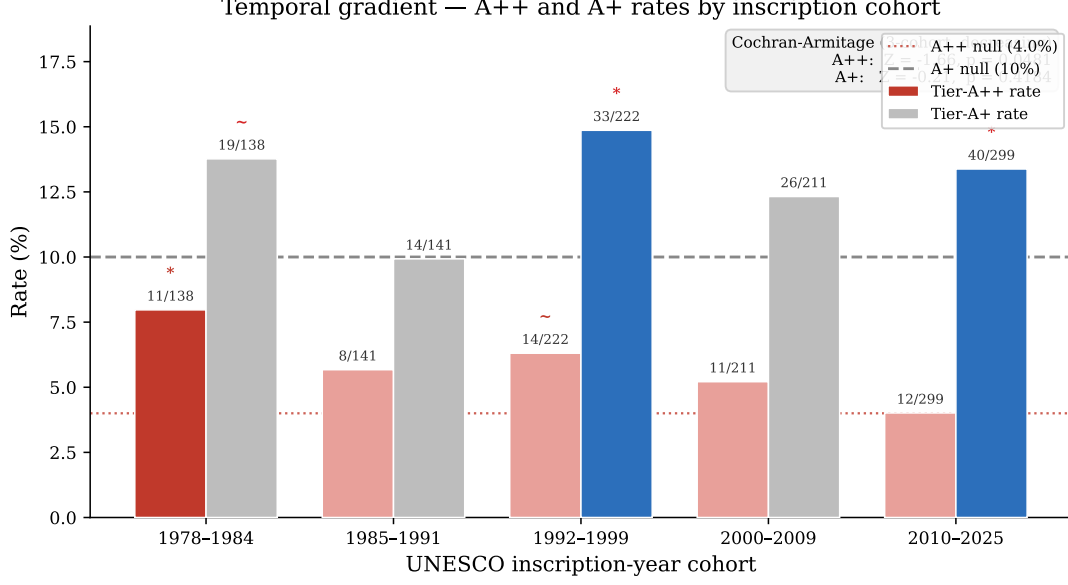


Figure 3: Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978–1984) shows an elevated A++ rate (8.0%, $p = 0.0232$ *) that declines monotonically across later cohorts toward the 4% null (dotted line; Cochran-Armitage $Z = -1.664$, $p = 0.0481$ *). The A+ rate (right bars) shows no significant trend ($p = 0.4184$ ns).

harmonic Mann-Whitney ($p_{\text{adj}} = 0.0012$); sequential-peak permutation ($p_{\text{adj}} = 0.0279$; §4.7); and evolution combined A++ (binomial $p = 0.0208$).

5 Robustness Checks

5.1 Unit Sweep and Sensitivity Slope

The unit sweep and sensitivity slope results are reported in §4.1 as part of the primary periodicity identification. Briefly, 3° is the uniquely identified period: adjacent non-harmonic spacings are non-significant, a $\pm 0.3\%$ shift collapses joint significance, and the Monte Carlo sweep (§6.2) independently confirms the peak. The 3° periodicity is also independently confirmed by Tests C and D (§4.3).

5.2 Global Anchor Sweep

A global sweep (0–360°E, 0.01° resolution, 36,000 trial anchors) assesses whether the $\sim 35^\circ\text{E}$ corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$), each anchor excluding its own entry, so the focal anchors test against $N = 1011$.

Both anchors place the corridor in the global top 2.3% (36,000 trial anchors, 0–360°E; Gerizim: 132 A+, 99.7th percentile; Jerusalem: 128 A+, 97.7th percentile, and the result is anchor-invariant within the 4.1km corridor window. No other historical meridian reaches significance.

Table 3: Multiple-comparisons summary. Primary family: Tests 1–3 (threshold-free, $k = 3$). Supporting enrichment tests (Tests 4–5) use a pre-specified $p_0 = 10\%$ proximity window; see §3.5.

Test	N	Raw p	Holm $p_{\text{adj}}, k=3$	Status
Test 1 (dome VM mix.)	90	0.00180 ^b ($w = 0.0876, \kappa = 252.19$)	0.004 **	Primary
Test 2 (Spearman)	1011	0.00065 ^b ($\rho = -0.1076$)	0.002 **	Primary
Test 3 (mean dev.)	1011	0.04729 ^b	0.047 *	Primary
Test 4 (full-corpus A+)	1011	0.0011 ^a	—	Supporting
Test 5 (dome A+)	90	0.0032 ^a (OR = 1.77)	—	Supporting
Test 6 (mound evol.)	110	0.0658 ^a (OR = 1.57)	N/A	Sensitivity
Test 7 (temporal)	1011	0.4184 ^b	—	Descriptive

^aBinomial or Fisher exact p vs. 10% geometric null (window = 0.15° ; convenience choice, sensitivity-audited over 2%–10% null rates in Supplementary S1); ^bpermutation or bootstrap p . Tests 1–3 are threshold-free; no tier cutoff enters any primary statistic. Holm (1979) step-down (?).

The $x.25^\circ\text{E}$ periodic structure

Every $x.25^\circ\text{E}$ longitude (120 anchors) produces identically the maximum A+ count, spaced 3° apart. The flat-top width is a function of the tier threshold chosen: A++ ($\pm 0.06^\circ$), A+ ($\pm 0.15^\circ$), A ($\pm 0.3^\circ$); the number of tied anchors is therefore a methodological artifact, not a physical feature. Mount Gerizim is a fixed landmark at 35.269°E (phase 2.269° , gap $0.019^\circ \approx 1.8 \text{ km}$); it lies within the A and A+ flat-top bands but not A++.

Formal statistical tests for the $x.25^\circ$ periodicity

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) confirms no spurious global periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0061$, **) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.0753$, †, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) confirm 3° as the dominant spacing among candidate values, *given* the pre-specified Gerizim anchor; neither sweep constitutes anchor-free or period-free discovery.

Corridor and landmark ranking. Of the 13 A+ sites unique to Gerizim and 9 unique to Jerusalem, 119 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #8 of 1012 (top 2.8%).

5.3 Spatial Autocorrelation

A+ harmonics host $4.24\times$ as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC ($\pm 0.5^\circ$, $\rho = 0.187$): DEFF = 1.842, $N_{\text{eff}} = 549$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [10.50%, 15.43%] excludes the 10% null. Within-group permutation using UNESCO-region $\times$ longitude bins reproduced the observed A+ count exactly in every trial, confirming this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.5) address that question directly.

5.4 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, [Good 2005](#)): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.060$ [†]; 1978-1984 canon ($N = 138$) $p = 0.433$ ns; pre-2000 ($N = 501$) $p = 0.490$ ns. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.070$ [†]. (3) KDE-smoothed null: expected 101.1 ± 9.6 , observed 132, $Z = 3.23$, $p = 0.001$ **. The KDE null ($p = 0.001$ **) confirms dome enrichment; permutation and bootstrap are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.004$ **, bootstrap $p = 0.006$ **). The geographic-concentration nulls (Null A, Null C; §4.5) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.5 Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) independently of the UNESCO corpus ([Tenenbaum, 2026](#)). The Kedu Plain (Java) hosts the densest cluster at the 75° harmonic (75°E from Gerizim), anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) ([Miksic, 1990](#)); the World Peace Pagoda at Lumbini achieves A++ precision (~ 555 m) uniquely from Gerizim.

Using the identical anchor and null, the corpus yields 70/229 Tier-A sites (30.6%, $1.53\times$ null; binomial $p < 0.001$, ***; permutation $Z = 1.59$, $p_{\text{perm}} = 0.065$, $\sim$). No Tier-A+ enrichment is found (binomial $p = 0.021$, *), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry $2.56\times$ more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.4). The Java/Sumatra node (105°–115°E, near the 75° harmonic at 110.3°E) concentrates 43 of 83 sites at Tier-A (51.8%, OR = $4.74\times$, $p < 0.001$ ***), rising to 42/54 (77.8%, OR = $18.38\times$, $p < 0.001$ ***) in the 107°–112°E window, where A+ is also elevated at 31.5% (OR = $4.57\times$, $p < 0.001$ ***). The physical heartland, India/Nepal (70°–90°E) and Myanmar/Cambodia (90°–105°E), is A-tier depleted (combined OR = $0.21\times$, $p < 0.001$ ***, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 35.8% (OR = $4.47\times$, $p < 0.001$ ***) and the combined heartland 29.2% (OR = $6.00\times$, $p < 0.001$ ***). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.4). Non-heartland Wikidata stupas (outside 70°–110°E, $N = 102$) show the same A-enriched, C-depleted profile: 50.0% reach Tier-A (OR = $5.68\times$, $p < 0.001$ ***) with no C-band enrichment (OR = $0.19\times$, $p = 1.000$ ns).

5.6 Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 99.7th percentile; a $\pm 0.3\%$ unit shift collapses joint significance. FDR audit (44 tests) retains 22 at $q < 0.05$; 6 survive Bonferroni at $k = 44$.

6 Limitations

6.1 Circularity: Full Canonical Treatment

Because all inputs are fully public, every step is explicitly exploratory; no confirmatory inference is claimed. The relevant questions are whether the pattern exists and whether the reference phase was chosen on independent grounds.

Both are answered directly. The global anchor sweep (§5.2) confirms that the enrichment is a property of the $\sim 35^\circ\text{E}$ corridor: every $x.25^\circ\text{E}$ anchor produces the same result, and Gerizim and Jerusalem both place the corridor in the global top 2.3% of 36,000 trial anchors. Gerizim is distinguished by pre-existing documentary evidence (the *tabbur ha-arets* cosmographic role, §2.2) predating this analysis.

The 3° structure is recoverable by any unanchored 120-phase scan. Dome enrichment (Test 2) is identical across all 120 $x.25^\circ\text{E}$ anchors. Cluster asymmetry (Test 3) requires no keyword filter and no anchor. Anchor-discovery correction (Test E, $p = 0.0753$, [†]) is marginal; primary evidence is Tests 2 and 3.

6.2 Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor of Cultural/Mixed sites. The regional distribution is as follows: (Europe & North America: 50%; Arab States: 9%; Asia-Pacific: 23%; Latin Americas: 11%; Africa: 7%). Domed monuments exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0393$, *), establishing that the enrichment is carried by architectural type, not location alone.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacings $0.5^\circ\text{--}15^\circ$) finds the dominant peak at 3° ($p < 0.001$), with harmonics at 6° and 12° also significant. The dome subset yields 1 independent dominant spacings ($P = 0.2248$ under the geographic null).

Periodogram and variable anchors. An omnidirectional Rayleigh periodogram finds the period with maximum phase concentration across all phases; the dominant period and optimal phase both vary by sub-population, so 3° /Gerizim is not the raw spectral peak in any single corpus. Its evidential weight rests on cross-corpus consistency: the UNESCO dome subset (Tests 1–3), Wikidata Q180987 stupas (§5.5), and the OWTRAD network (§7.2) all show enrichment at the same pre-specified period and anchor.

6.3 Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows A+ rate 9.7%, below the 10% null ($p = 0.5149$ ns, one-sided depletion).

6.4 Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids (unsystematic noise, no directional bias).

Metrological evidence. No ancient surveying instrument calibrated to metrological sub-units has been recovered; the *bēru* (Hunger & Steele, 2019; Powell, 1987) is noted as historical coincidence. All thresholds are degree-denominated.

7 Discussion

7.1 Alternative Explanations

(1) **Independent convergence.** Hemispherical structures recur across cultures (Aveni, 2001, 2003), but convergence predicts no longitude specificity.

(2) **Geographic concentration.** Null C: other monuments from the same longitude footprint are significantly less enriched ($p = 0.0393$, *); the Monte Carlo unit sweep (§6.2) confirms 3° as the dominant spacing given the pre-specified Gerizim anchor, ruling out the explanation that the enrichment reflects only the concentration of cultural sites in the 35°E longitude corridor.

(3) **Random coincidence.** The KDE-smoothed null confirms dome enrichment ($p = 0.001$ **); permutation ($p = 0.060$ $\dagger$) and bootstrap ($p = 0.070$ $\dagger$) are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.004$ **; bootstrap $p = 0.006$ **). Tests C and D (§4.3) confirm anchor specificity and periodicity structure.

7.2 Independent Validation: OWTRAD and Wikidata Corpora

The 3° period identified in the UNESCO corpus is independently confirmed in two external datasets. First, the OWTRAD attested-route network (Ciolek, 2004) (1946 route segments, 1674 unique endpoints). Testing all 3498 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 193 fall at A++ harmonics, 420 at A+, and 791 at A ($1.38\times$ null rate, $z = 4.58$, $p = < 0.0001$ *** for A++; $p = < 0.0001$ *** for A+; $p = < 0.0001$ *** for A; one-tailed binomial). Testing each unique endpoint once gives 84/1674 at A++, 190 at A+, and 362 at A ($1.25\times$, $z = 2.13$, $p = 0.0224$ * for A++; $p = 0.0377$ * for A+; $p = 0.0524$ $\dagger$ for A; one-tailed binomial). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.0202$ *, confirming the enrichment is specific to the Gerizim anchor. A cluster asymmetry test mirrors Test 3 of the primary family: harmonics containing at least one A++ endpoint (30 harmonics) carry $2.70\times$ more total route-connectivity than harmonics without one (11 harmonics; $p = 0.0007$ ***, Mann-Whitney one-tailed). The Silk Road corridor is illustrated in Figure 5; harmonic tier distributions for both OWTRAD populations are shown in Figure 4.

8 Conclusion

The UNESCO Cultural and Mixed corpus shows significant 3° longitudinal enrichment with all three primary tests surviving Bonferroni correction at $k = 3$. Longitude bands containing a precisely aligned site are $4.24\times$ more densely populated than bands without one. A multi-scale combined test across 15 log-spaced proximity windows confirms the result is not an artifact of any fixed tier choice (dome subset $p = 0.00362$ **); joint enrichment across four independent corpora peaks at approximately 29 km from a harmonic node ($p = 0.01232$ *). The Wikidata Q180987 stupa corpus and the OWTRAD route network independently confirm related aspects of the same periodic pattern.

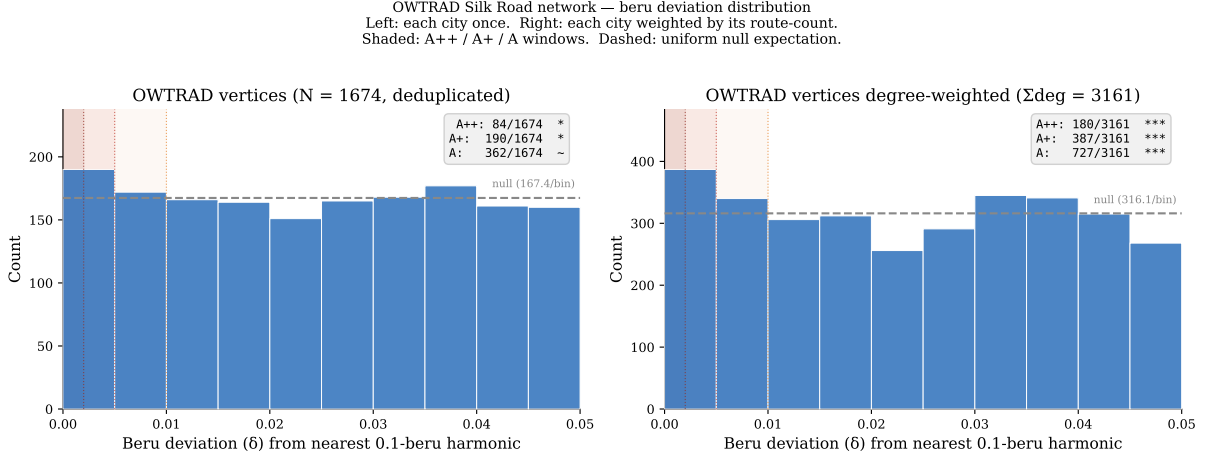


Figure 4: OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3498 degree-weighted positions (right). The degree-weighted distribution shows significant A++ enrichment ($z = 4.58$, $p < 0.0001$ ***); the deduplicated endpoint distribution does not ($z = 2.13$, $p = 0.0224$ *).

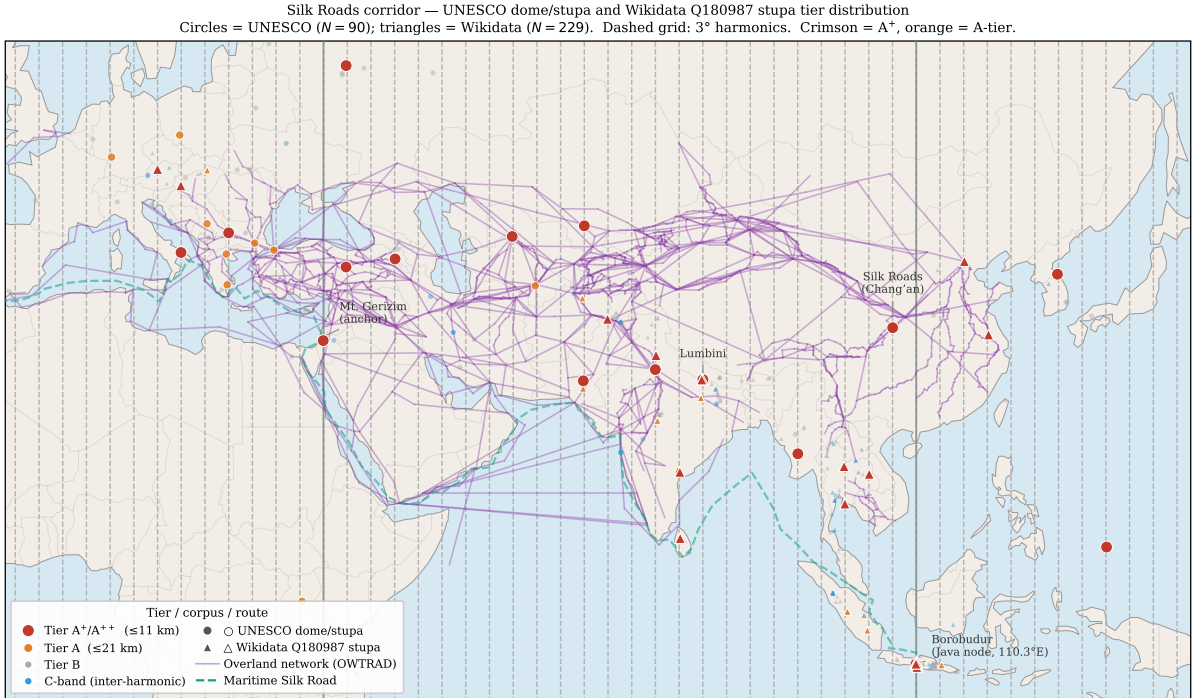


Figure 5: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek, 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (supplementary/interactive_map/silkroad_interactive.html).

A Von Mises Mixture Characterisation of External Corpora

A von Mises + uniform mixture model $f(\theta|\kappa) = w \cdot \text{VM}(0, \kappa) + (1-w) \cdot \text{Uniform}(0, 2\pi)$ was fitted to the harmonic phase deviations ($\theta \in [0, \pi]$, $\theta = 0$ on-harmonic) of two

external corpora using maximum likelihood with a grid of starting values.

OWTRAD Silk Road nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = \kappa^{-1/2} = 0.1543^\circ$.

Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922^\circ$.

N -weighted pooled $\hat{\sigma} = 0.1588^\circ$; κ ratio = 1.6 across corpora.

These numbers characterize the angular clustering of the external corpora. They are not used to derive the UNESCO tier boundaries or any inferential threshold in this paper. The $\hat{\sigma}$ multiples ($0.25\hat{\sigma} = 0.048^\circ$, $0.50\hat{\sigma} = 0.096^\circ$, $1.00\hat{\sigma} = 0.1923^\circ$) agree with the tier widths to within rounding (Table ??), which is noted as descriptive context only.

B Tier-Threshold Sensitivity Audit

The primary enrichment tests use Tier-A+ ($P_0 = 4\%$, $\delta \leq 0.060^\circ$, ≈ 7 km). To confirm that this choice does not manufacture the signal, we swept the null-rate parameter from 1% to 10% in 0.5% steps, recomputing the threshold as $\delta_{\max} = P_0 \times 1.5^\circ$ and re-running the binomial one-sided test against each of eight subsets: full corpus, religion-founding sites, dome/tholos/stupa (UNESCO), dome + mound, founding sites (any category), UNESCO stupa keyword, Wikidata Q180987 stupa corpus, and the Java-region stupa sub-corpus. All six tier levels (A++, A+, A, C, C-, C-) are evaluated at the primary null rates (A++/C- = 2%; A+/C- = 4%; A/C = 8%).

Across the full-corpus sweep, odds ratios are consistently positive (OR = 1.09–1.43) and binomial $p < 0.05$ for all null rates $\geq 3.5\%$. Sub-corpus results are consistent: the dome/tholos/stupa subset shows OR = 2.2–4.0 with $p < 0.05$ across the entire 1%–10% range; founding sites OR = 1.4–2.4; religion-founding sites OR = 1.3–1.7. C-tier (anti-harmonic) odds ratios are uniformly close to 1.0 and non-significant, confirming that the enrichment is specific to harmonic proximity and not an artifact of the binomial test or bin-width choice.

Full results are in `supplementary/audit/tier_sensitivity_audit.txt` (with machine-readable CSVs `tier_sensitivity_part1.csv` and `tier_sensitivity_part2.csv`); regenerate with `python3 tools/generate_audit_tier_sensitivity.py` or `bash tools/update_all_audits.sh`.

C Keyword Audit Summary

Full site-by-site audit files are indexed in Supplementary S1 (`supplementary/Supplementary_S1.md`; [Tenenbaum 2026](#)); regenerate all with `bash tools/update_all_audits.sh`.

C.1 Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 1bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating.

For Test 5x, ambiguous keywords (*dome*, *domed*, *domes*, *spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected

sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuta, and Diquís Stone Spheres.

Test 6 adds earthen mound-stage keywords (*tumulus*, *tumuli*, *barrow*, *barrows*, *kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 6x applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.5 and §4.6.

C.2 Founding-Origin Classifier (Test 5, Post-Hoc)

Results. The classifier finds 31.8% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.19, $p = 0.225$ ns). The A++ sub-population (56 sites) shows a stronger signal: 42.9% carry a founding keyword (OR = 1.94, $p = 0.014$ *), consistent with the most precisely aligned sites being disproportionately historically foundational monuments.

The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum, 2026).

References

- Aveni, A. F. (2001). *Skywatchers: A Revised and Updated Version of Skywatchers of Ancient Mexico*. University of Texas Press.
- Aveni, A. F. (2003). Archaeoastronomy in the ancient Americas. *Journal of Archaeological Research*, 11(2), 149–191.
- Casson, L. (1989). *The Periplus Maris Erythraei: Text with Introduction, Translation, and Commentary*. Princeton University Press.
- Ciolek, T. M. (2004). *Old World Trade Routes (OWTRAD) Project*. Asia Pacific Research Online, Canberra. <http://www.ciolek.com/OWTRAD/DATA/tmcOWTRAD.html>
- Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.
- Crown, A. D. (1989). *The Samaritans*. J. C. B. Mohr (Paul Siebeck).
- Freeman, P. R. (1976). A Bayesian analysis of the megalithic yard. *Journal of the Royal Statistical Society, Series A*, 139(1), 20–35.
- Good, P. I. (2005). *Permutation, Parametric, and Bootstrap Tests of Hypotheses* (3rd ed.). Springer.
- Hansen, V. (2012). *The Silk Road: A New History*. Oxford University Press.
- Harvey, P. D. A. (1987). Medieval maps of the world. In J. B. Harley & D. Woodward (Eds.), *The History of Cartography, Volume 1: Cartography in Prehistoric, Ancient, and Medieval Europe and the Mediterranean* (pp. 283–370). University of Chicago Press.

- Burton, R. F. (1893). *Personal Narrative of a Pilgrimage to Al-Madinah & Meccah*, vol. 2 (Memorial Edition), p. 63, n. 2. Tylston and Edwards.
- Hunger, H. & Steele, J. (2019). *The Babylonian Astronomical Compendium MUL.APIN*. Routledge.
- Thureau-Dangin, F. (1921). *Numération et métrologie sumériennes*. *Revue d'Assyriologie et d'archéologie orientale*, 18, 123–142.
- Kendall, D. G. (1974). Hunting quanta. *Philosophical Transactions of the Royal Society A*, 276(1257), 231–266.
- MacDonald, J. (1964). *The Theology of the Samaritans*. SCM Press.
- Mardia, K. V. & Jupp, P. E. (2000). *Directional Statistics*. Wiley.
- Magli, G. (2013). *Architecture, Astronomy and Sacred Landscape in Ancient Egypt*. Cambridge University Press.
- Magli, G. (2016). *Archaeoastronomy: Introduction to the Science of Stars and Stones*. Springer.
- Miksic, J. N. (1990). *Borobudur: Golden Tales of the Buddhas*. Periplus Editions.
- Neugebauer, O. (1975). *A History of Ancient Mathematical Astronomy*, 3 vols. Springer.
- Powell, M. A. (1987). Masse und Gewichte. In D. O. Edzard (Ed.), *Reallexikon der Assyriologie und Vorderasiatischen Archäologie*, vol. 7, pp. 457–517. Walter de Gruyter.
- Purvis, J. D. (1968). *The Samaritan Pentateuch and the Origin of the Samaritan Sect*. Harvard University Press.
- Robson, E. (2008). *Mathematics in Ancient Iraq: A Social History*. Princeton University Press.
- Ruggles, C. L. N. (1999). *Astronomy in Prehistoric Britain and Ireland*. Yale University Press.
- Ruggles, C. L. N. & Cotte, M. (Eds.) (2010). *Heritage Sites of Astronomy and Archaeoastronomy in the Context of the UNESCO World Heritage Convention: A Thematic Study*. ICOMOS & International Astronomical Union.
- Ruggles, C. L. N. (Ed.) (2015). *Handbook of Archaeoastronomy and Ethnoastronomy*. Springer.
- Sen, T. (2014). The spread of Buddhism. In B. Z. Kedar & M. E. Wiesner-Hanks (Eds.), *The Cambridge World History, Vol. 5: Expanding Webs of Exchange and Conflict, 500 CE–1500 CE* (pp. 447–479). Cambridge University Press.
- Silva, F. (2020). A probabilistic framework and significance test for the analysis of structural orientations in skyscape archaeology. *Journal of Archaeological Science*, 118, 105138.

- Tenenbaum, S. (2026). *Longitude Quantization in the UNESCO World Heritage Corpus: Domes, Stupas, and 3° Harmonics: Code and Data* (v1.6.0). Zenodo. <https://doi.org/10.5281/zenodo.19938283>
- Tomber, R. (2008). *Indo-Roman Trade: From Pots to Pepper*. Duckworth.
- Thom, A. (1964). The larger units of length of megalithic man. *Journal of the Royal Statistical Society, Series A*, 127(4), 527–533.
- Thom, A. (1967). *Megalithic Sites in Britain*. Oxford University Press.
- Thom, A. (1971). *Megalithic Lunar Observatories*. Oxford University Press.
- UNESCO World Heritage Centre. (2024). *World Heritage List* [XML dataset]. Retrieved from <https://whc.unesco.org/en/list/xml/>.
- Watson, R. T. & Horowitz, W. (2011). *Writing Science Before the Greeks: A Naturalistic Analysis of the Babylonian Astronomical Treatise MUL.APIN*. Brill.
- Williamson, T. & Bellamy, L. (1983). *Ley Lines in Question*. World’s Work.

Data Availability Statement

All data and code are openly available; a permanent snapshot (v1.6.0) is deposited at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>); UNESCO XML: UNESCO World Heritage Centre 2024. Scripts, audit files, and maps are indexed in Supplementary S1 (supplementary/Supplementary_S1.md), with each file’s role, macros produced, and paper section noted. Full reproduction: `reproduce_all_macros.sh`, `update_all_audits.sh`, `generate_figures.py`.

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Conflict of Interest

The author declares no conflict of interest.

Software and AI Disclosure

All statistical analyses were performed in Python 3 using NumPy, SciPy, and pandas. Large-language-model tools were used during preparation of this manuscript: Claude (Anthropic) assisted with prose drafting and editing; GitHub Copilot (Microsoft) assisted with code autocompletion. These tools were used solely as writing and coding aids. All scientific hypotheses, analytical choices, keyword-list decisions, interpretation of results, and conclusions are solely the responsibility of the author. No AI tool was used to generate, fabricate, or substitute for primary data or statistical analysis.